

Single-Factor Between-Participants ANOVA

PSYC5141: Experimental Methods for Psychology

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Week 2, Lecture 1

Learning Objectives

- Introduction to Analysis of Variance (ANOVA)
- Sources of variability in raw data
- The F ratio for a between-participants design
- Calculating F ratios for between-participants designs
- Calculating the between-group variance
- Calculating the within-group variance
- Producing the F statistic

Introduction To ANOVA

Analysis of Variance

- **Analysis of Variance (ANOVA)** is a collection of statistical techniques for examining group differences
- ANOVA extends the methods covered so far to more complex experimental designs:
 - a single **factor** (i.e., independent variable) with *three or more levels* (**single-factor designs**, Weeks 1-4; Mark)
 - *two or more factors* within the same design (**factorial designs**, Weeks 5-8; Emma)
- **Today:** single-factor between-participants ANOVA

Limitations Of The t -Test

A t-test allows us to compare two means

- Many experiments involve more than two treatment groups
- Multiple t-tests could be used to compare the groups ...
- ... but conducting multiple tests inflates the probability of a Type I error
- **ANOVA allows us to test for differences among multiple means simultaneously**

Sources Of Variability In Raw Data

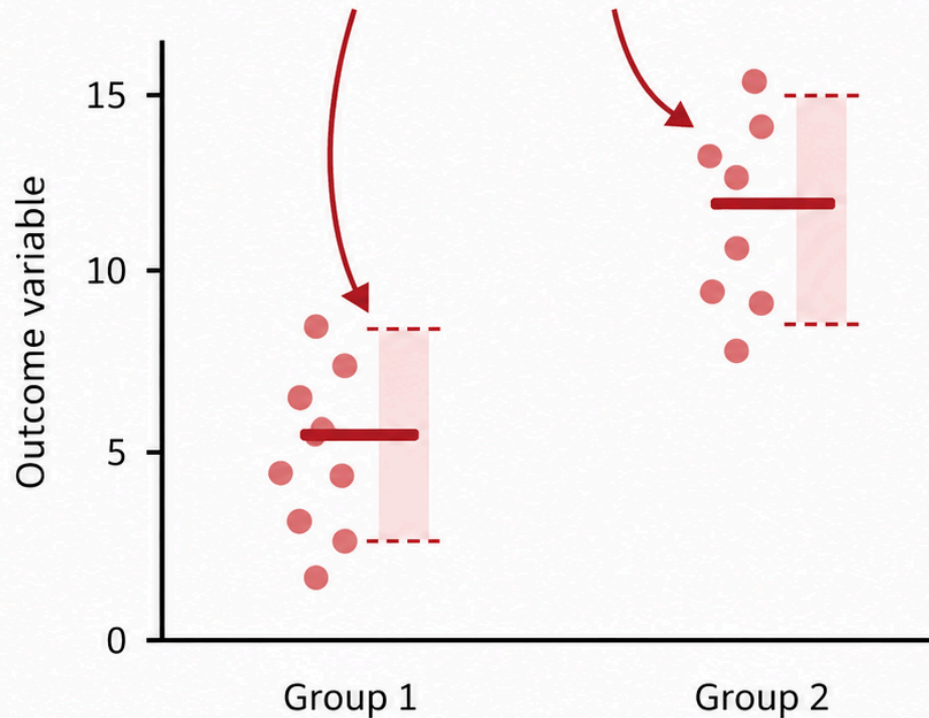
Recap: Sources of Variability in Between-Participants Designs

- In a between-participants experiment, scores vary between groups and within groups
- **Between-group variability:** refers to differences between the group means
- **Within-group variability:** refers to differences among scores within each group
- Detecting a treatment effect depends on the size of the **between-group differences relative to the within-group variability**

Visual Depiction of Between- and Within-Group Variability

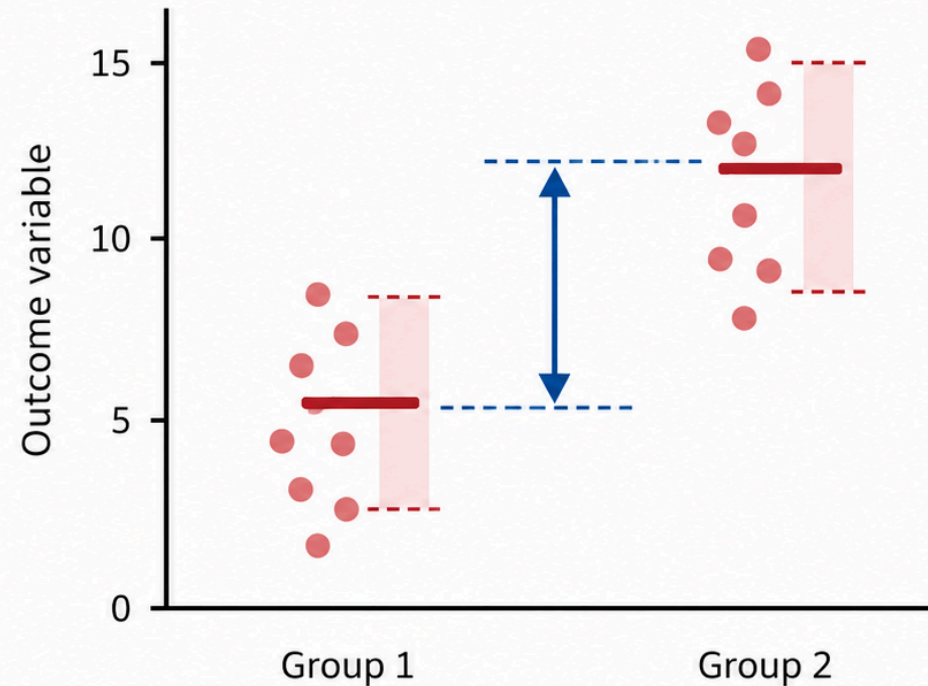
Within-group variability

Variation in scores among individuals within the same group



Between-group variability

Differences in the group means across groups



Examples of Between- and Within-Group Variability

	High between-group variability, no within-group variability			No between-group variability, high within-group variability			Moderate between-group variability, moderate within-group variability		
	Group 1	Group 2	Group 3	Group 1	Group 2	Group 3	Group 1	Group 2	Group 3
	10	20	30	10	15	5	10	10	20
	10	20	30	25	20	25	10	20	20
	10	20	30	30	30	25	10	20	30
	10	20	30	35	40	45	20	20	30
	10	20	30	50	45	50	20	30	30
Mean	10	20	30	30	30	30	14	20	26
<i>S</i>	0	0	0	14.6	12.8	18.0	5.5	7.1	5.5

What *Causes* This Variability?

Scores in an experiment can vary for three main reasons:

1. Treatment effects
2. Individual differences
3. Random (residual) errors

Sources of Variability

Scores in an experiment can vary for three main reasons:

1. **Treatment effects**
2. Individual differences
3. Random (residual) errors

Treatment effects → Between-Group Variability

- If the independent variable has an effect, the treatment conditions should produce different scores
- These differences should be reflected in **different group means**
- Treatment effects therefore contribute to **between-group variability**
- **Caveat:** between-group variability is not necessarily *only* treatment variability

Sources of Variability

Scores in an experiment can vary for three main reasons:

1. **Treatment effects**
2. Individual differences
3. Random (residual) errors

Sources of Variability

Scores in an experiment can vary for three main reasons:

1. Treatment effects
2. **Individual differences**
3. Random (residual) errors

Individual Differences → Primarily Within-Group Variability

- Participants naturally differ in characteristics that influence their scores
- Because different participants make up each treatment condition, their scores will vary even when they receive the **same treatment**
- Individual differences therefore contribute to **within-group variability**
- Large individual differences can make a treatment effect harder to detect

Sources of Variability

Scores in an experiment can vary for three main reasons:

1. Treatment effects
2. **Individual differences**
3. Random (residual) errors

Sources of Variability

Scores in an experiment can vary for three main reasons:

1. Treatment effects
2. Individual differences
3. **Random (residual) errors**

Random (Residual) Errors → Primarily Within-Group Variability

- Scores can also vary because of unpredictable factors unrelated to either the treatment or stable individual differences
- Examples might include:
 - varying external conditions (e.g., time of day, temperature)
 - state of the participant (e.g., current focus of attention)
 - experimenter's ability to measure and score accurately
- Because these factors can affect participants **within the same treatment condition**, they contribute to **within-group variability**

...Experimental Errors

- Collectively, individual differences and random errors are known as **experimental errors**
- The aim of every experiment is to minimise experimental errors, so they do not obscure treatment effects
- Such errors spread data away from the group means, increasing the variability in scores and their standard deviations
- **Larger standard deviations = more experimental error**

The F Ratio For A Between-Participants Design

The Logic of ANOVA

- ANOVA evaluates treatment effects by comparing:
 - the variability **between groups**
 - with the **experimental error**
- If the treatment has an effect, we expect the variability between groups to be relatively large compared with experimental error

$$F = \frac{\text{between-group variance}}{\text{experimental error}}$$

For A Between-Participants ANOVA

$$F = \frac{\text{between-group variance}}{\text{within-group variance}}$$

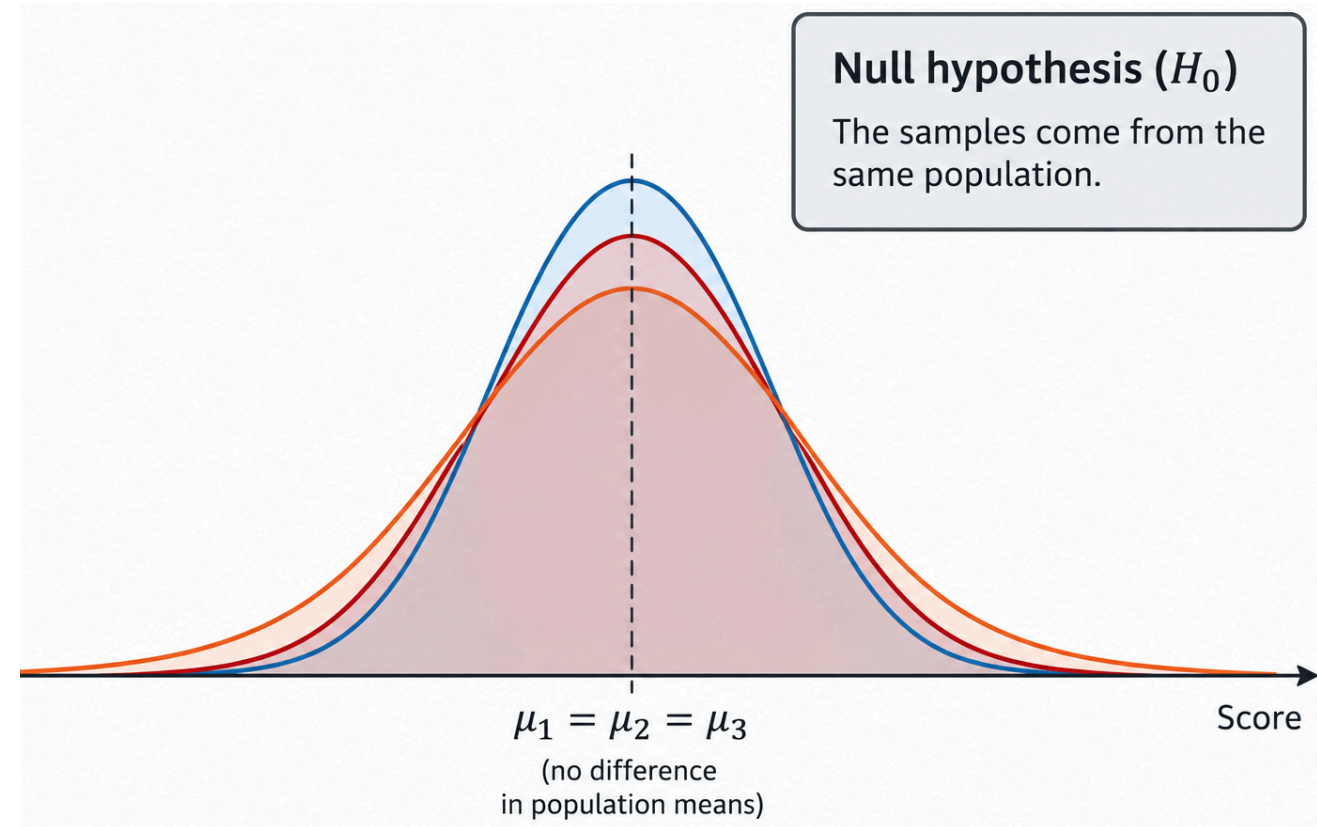
$$F = \frac{\text{treatment effects} + \text{experimental error}}{\text{experimental error}}$$

$$F = \frac{\text{treatment effects} + \text{individual differences} + \text{random errors}}{\text{individual differences} + \text{random errors}}$$

Testing For Differences

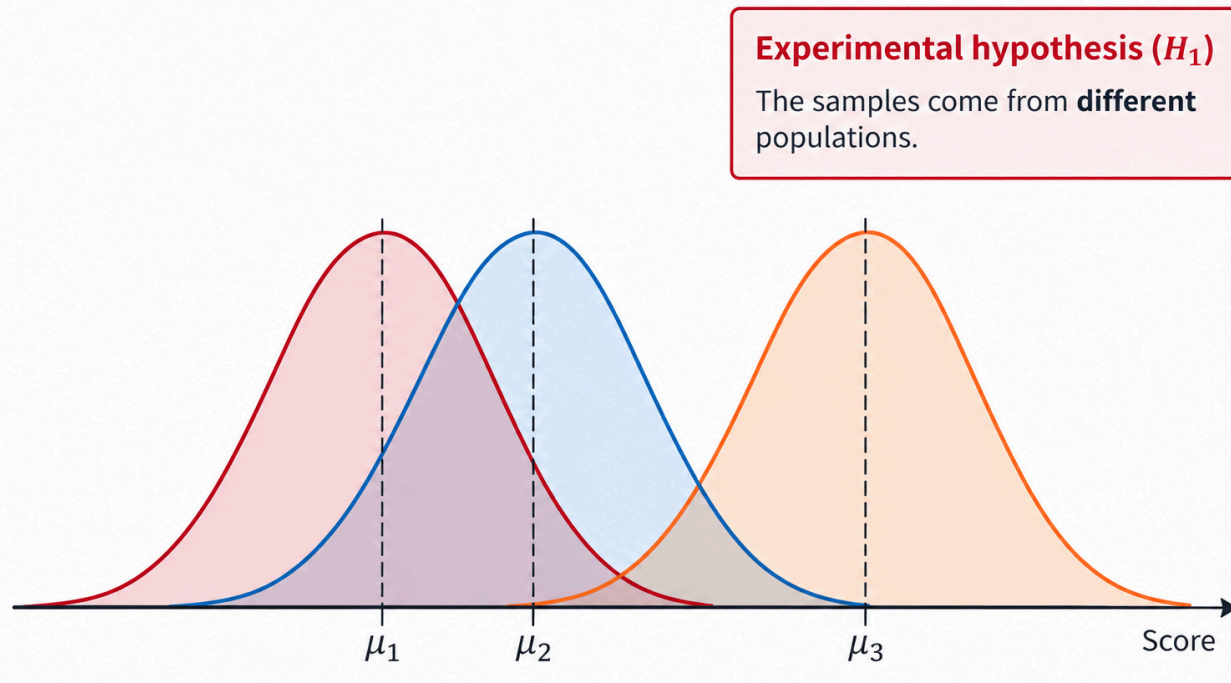
H_0 the Null Hypothesis

- Under H_0 , the samples come from the **same** population
- $\mu_1 = \mu_2 = \mu_3$ [No difference in the population means]
- Experimental effect = 0
- All differences are due to individual differences + random (residual) errors



Testing For Differences

H_1 the Alternative Hypothesis



- Under H_1 , the samples come from **different** populations
- At least one population mean differs
- Experimental effect > 0
- Differences are due to individual differences, random (residual) errors **and** the treatment effects

If The Null Hypothesis Is True...

- Under H_0 , the treatment effect is zero
- Between-group and within-group variability therefore reflect the same experimental error

$$F \approx \frac{0 + \text{experimental error}}{\text{experimental error}} \approx 1$$

- An F -ratio close to 1 is what we would expect if H_0 were true

If The Alternative Hypothesis Is True...

- Under H_1 , the treatment contributes additional between-group variability:

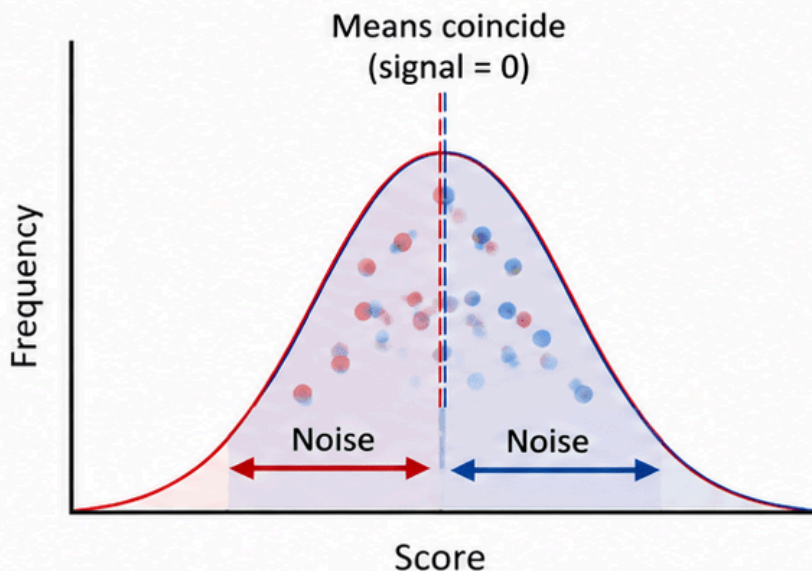
$$F = \frac{\text{treatment effects} + \text{experimental error}}{\text{experimental error}} > 1$$

- As the treatment effect increases relative to experimental error, the F -ratio becomes larger
- Overall, when looking for a treatment effect, you are looking for a relatively large F value

Signal vs. Noise: Understanding the F Ratio

No treatment effect

Group means are the same
(no treatment signal)

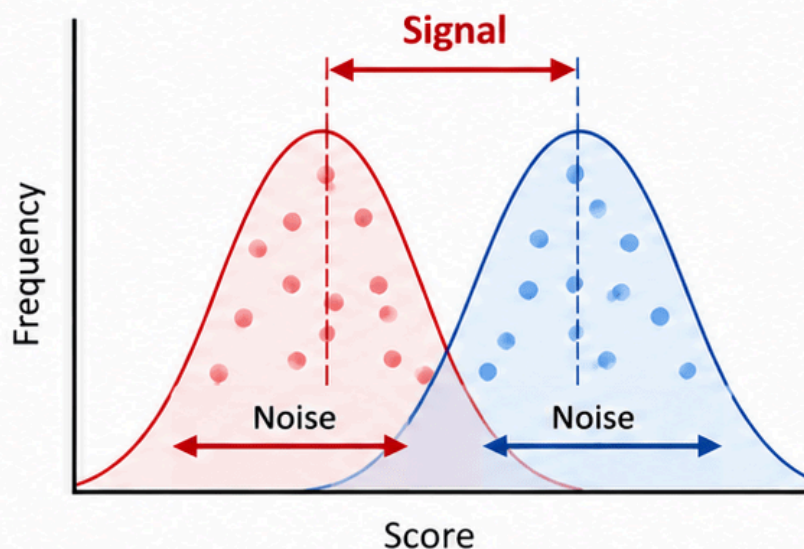


$F \approx 1$

(no treatment signal)

Moderate treatment effect

Group means are separated
(treatment signal > 0)

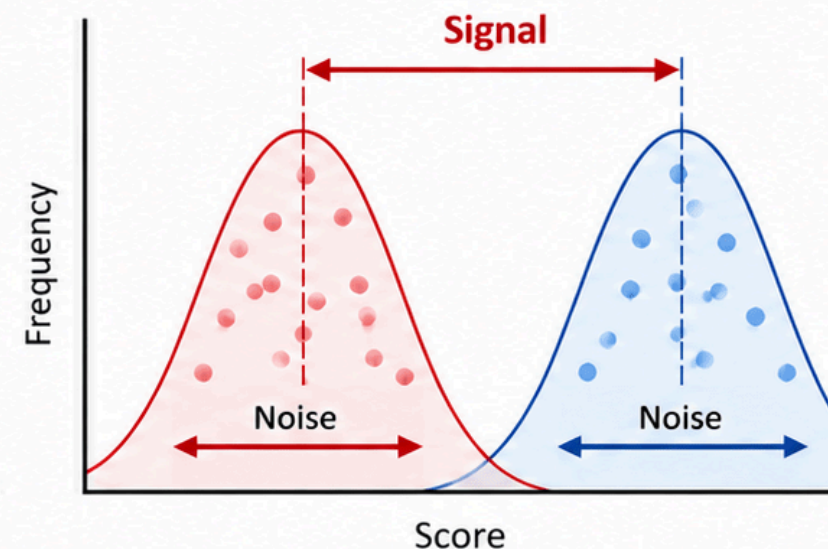


$F > 1$

(signal emerging from noise)

Large treatment effect

Group means are well separated
(large treatment signal)



Large F

(strong signal relative to noise)

Calculating F Ratios For Single-Factor Between-Participants Designs

Worked Example

Does study strategy affect test performance?

- The **factor** is study strategy with three **levels**:
 - A_1 : **Re-reading**: repeatedly reading the study material
 - A_2 : **Highlighting**: highlighting key information while studying
 - A_3 : **Retrieval practice**: actively recalling the material from memory
- Participants are randomly allocated to one of the three levels
- After studying, all participants complete the same test and obtain a test score
- **Research question**: *do mean test scores differ between the three study strategies?*

Worked Example



A_1 scores	A_2 scores	A_3 scores
5	8	10
7	8	11
8	10	12
9	11	13
11	13	14

Calculating The F Ratio

- To calculate the F ratio, we need two estimates of variance:

$$F = \frac{\text{total between-group variance}}{\text{total within-group variance}}$$

So, we will calculate:

1. Total between-group variance

How much variability is there between our groups?

2. Total within-group variance

How much variability is there within our groups?

Calculating The F Ratio

- To calculate the F ratio, we need two estimates of variance:

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So, we will calculate:

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How much variability is there between our groups?

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How much variability is there within our groups?

Calculating The Between-Group Variance

Calculating The Level Means



A_1 scores	A_2 scores	A_3 scores
5	8	10
7	8	11
8	10	12
9	11	13
11	13	14

Total of the set of scores

$$\overline{X} = \frac{\sum X}{N}$$

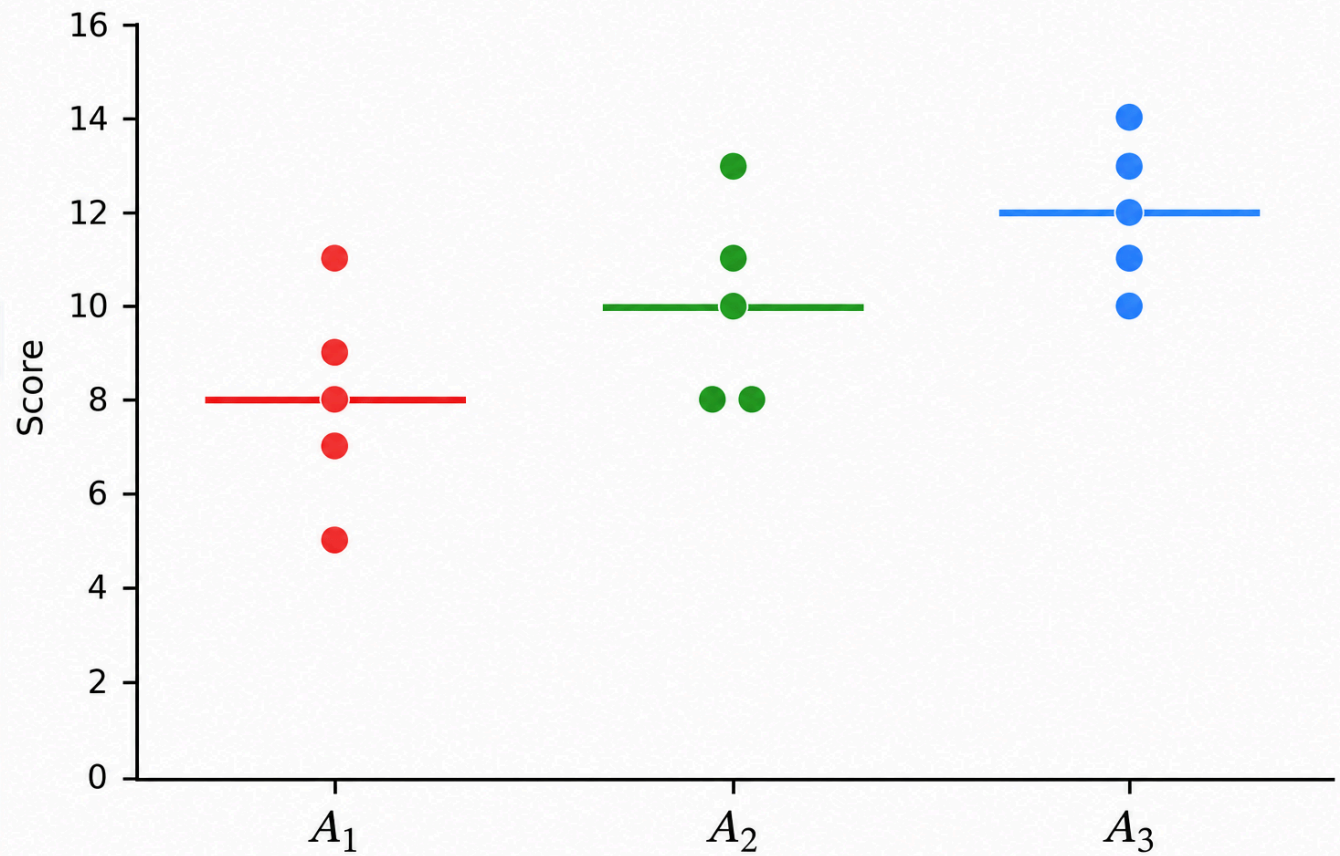
Mean

Number of scores

Calculating The Level Means



A_1 scores	A_2 scores	A_3 scores
5	8	10
7	8	11
8	10	12
9	11	13
11	13	14
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$



Calculating The Grand Mean



A_1 scores	A_2 scores	A_3 scores
5	8	10
7	8	11
8	10	12
9	11	13
11	13	14
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$

Grand mean

$$\bar{Y} = \frac{\bar{A}_1 + \bar{A}_2 + \bar{A}_3 + \dots \bar{A}_k}{k}$$

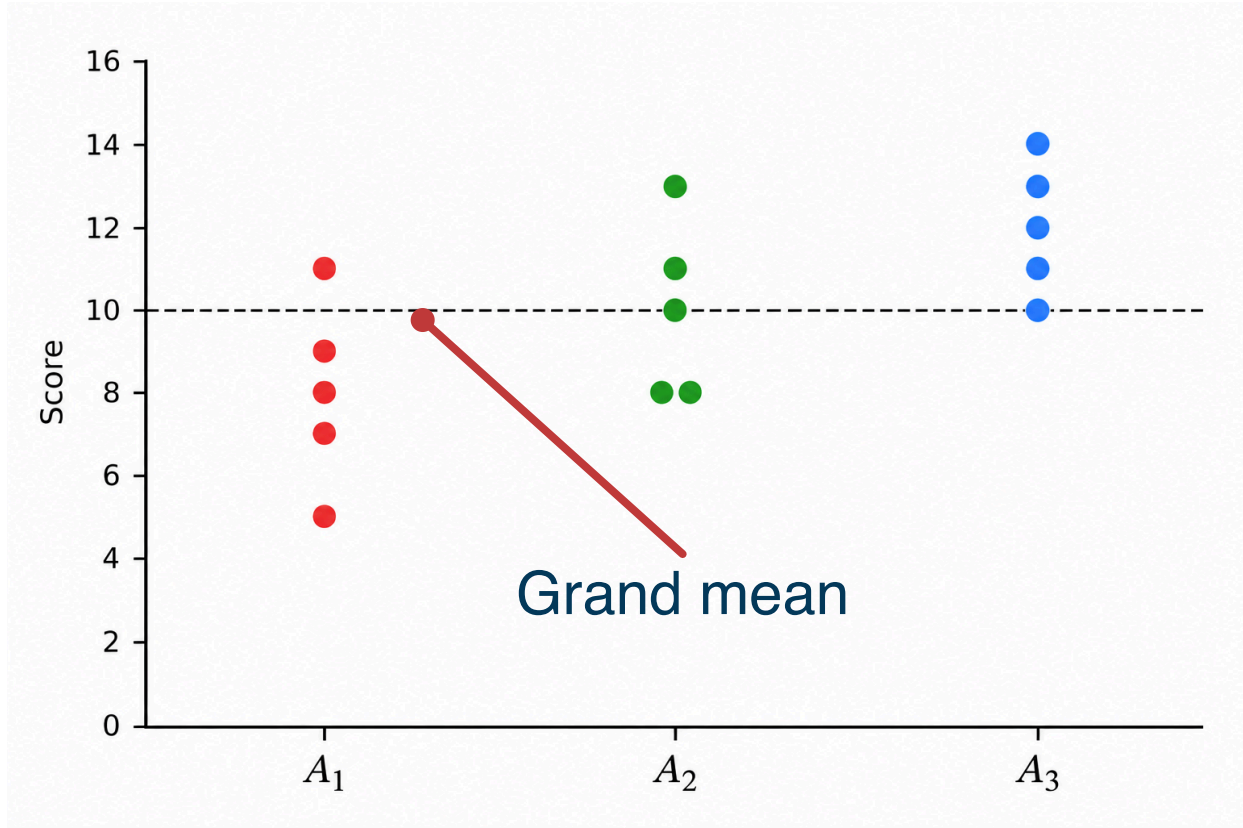
Number of levels

$$\bar{Y} = \frac{8 + 10 + 12}{3}$$

$$\bar{Y} = 10$$

Calculating The Grand Mean

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$



Total Between-Group Variance

total between-group variance = $\frac{\text{between-group sum of squares}}{\text{total between-group degrees of freedom}}$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance =
$$\frac{N_{A1}(\bar{A}_1 - \bar{Y})^2 + N_{A2}(\bar{A}_2 - \bar{Y})^2 + N_{A3}(\bar{A}_3 - \bar{Y})^2 (\text{and so on})}{\text{total between-group degrees of freedom}}$$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
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8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance =
$$\frac{N_{A1}(\bar{A}_1 - \bar{Y})^2 + N_{A2}(\bar{A}_2 - \bar{Y})^2 + N_{A3}(\bar{A}_3 - \bar{Y})^2 (\text{and so on})}{\text{total between-group degrees of freedom}}$$

			
A ₁ scores	A ₂ scores	A ₃ scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance =
$$\frac{N_{A1}(\bar{A}_1 - \bar{Y})^2 + N_{A2}(\bar{A}_2 - \bar{Y})^2 + N_{A3}(\bar{A}_3 - \bar{Y})^2 (\text{and so on})}{\text{total between-group degrees of freedom}}$$

			
A ₁ scores	A ₂ scores	A ₃ scores	
5	8	10	
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$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance =
$$\frac{N_{A1}(\bar{A}_1 - \bar{Y})^2 + N_{A2}(\bar{A}_2 - \bar{Y})^2 + N_{A3}(\bar{A}_3 - \bar{Y})^2 (\text{and so on})}{\text{total between-group degrees of freedom}}$$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
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$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance =
$$\frac{N_{A1}(\bar{A}_1 - \bar{Y})^2 + N_{A2}(\bar{A}_2 - \bar{Y})^2 + N_{A3}(\bar{A}_3 - \bar{Y})^2 (\text{and so on})}{\text{total between-group degrees of freedom}}$$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
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8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance = $\frac{5(8 - 10)^2 + 5(10 - 10)^2 + 5(12 - 10)^2}{\text{total between-group degrees of freedom}}$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance = $\frac{5(-2)^2 + 5(0)^2 + 5(2)^2}{\text{total between-group degrees of freedom}}$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance = $\frac{5(4) + 5(0) + 5(4)}{\text{total between-group degrees of freedom}}$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance = $\frac{20 + 0 + 20}{\text{total between-group degrees of freedom}}$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
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Total Between-Group Variance

40

total between-group variance =

total between-group degrees of freedom

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
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11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Degrees Of Freedom

Between-group degrees of freedom

- For the between-group variance, the relevant values are the **level means**
- The level means are subject to a constraint imposed by the **grand mean**
- Once all but one of the level means are known, the final level mean is determined
- Thus, the number of level means that are free to vary is:

$$\text{total between-group degrees of freedom} = \text{number of levels} - 1$$

- For our experiment, there are three levels:

$$\text{total between-group degrees of freedom} = 3 - 1 = 2$$

Total Between-Group Variance

40

total between-group variance =

total between-group degrees of freedom

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
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11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance = $\frac{40}{2}$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Between-Group Variance

total between-group variance = $\frac{40}{2} = 20$

N_{A1} = Number of scores for A_1
= 5

N_{A2} = Number of scores for A_2
= 5

N_{A3} = Number of scores for A_3
= 5

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
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9	11	13	
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$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Calculating The Within-Group Variance

Calculating The F Ratio

- To calculate the F ratio, we need two estimates of variance:

$$F = \frac{\text{total between-group variance}}{\text{total within-group variance}}$$

So, we will calculate:

1. Total between-group variance

How much variability is there between our groups?

2. Total within-group variance

How much variability is there within our groups?

Calculating The F Ratio

- To calculate the F ratio, we need two estimates of variance:

$$F = \frac{\text{total between-group variance}}{\text{total within-group variance}}$$

So, we will calculate:

1. Total between-group variance

How much variability is there between our groups?

2. Total within-group variance

How much variability is there within our groups?

Total Within-Group Variance

$$\text{total within-group variance} = \frac{\text{within-group sum of squares}}{\text{total within-group degrees of freedom}}$$

Total Within-Group Variance

$$\text{total within-group variance} = \frac{\sum(A_1 - \bar{A}_1)^2 + \sum(A_2 - \bar{A}_2)^2 + \sum(A_3 - \bar{A}_3)^2 + \dots}{\text{total within-group degrees of freedom}}$$

Total Within-Group Variance

total within-group variance =
$$\frac{\sum(A_1 - \bar{A}_1)^2 + \sum(A_2 - \bar{A}_2)^2 + \sum(A_3 - \bar{A}_3)^2 + \dots}{\text{total within-group degrees of freedom}}$$



Total Within-Group Variance

total within-group variance =
$$\frac{\sum(A_1 - \bar{A}_1)^2 + \sum(A_2 - \bar{A}_2)^2 + \sum(A_3 - \bar{A}_3)^2 + \dots}{\text{total within-group degrees of freedom}}$$

- Sum of squares for level A_1
- Sum of squares for level A_2
- Sum of squares for level A_3



A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Within-Group Variance

total within-group variance =
$$\frac{\sum(A_1 - \bar{A}_1)^2 + \sum(A_2 - \bar{A}_2)^2 + \sum(A_3 - \bar{A}_3)^2 + \dots}{\text{total within-group degrees of freedom}}$$

			
A ₁ scores	A ₂ scores	A ₃ scores	
5	8	10	
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8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Within-Group Variance

total within-group variance =
$$\frac{\sum(A_1 - 8)^2 + \sum(A_2 - 10)^2 + \sum(A_3 - 12)^2 + \dots}{\text{total within-group degrees of freedom}}$$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Within-Group Variance

total within-group variance = $\frac{48}{\text{total within-group degrees of freedom}}$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Degrees Of Freedom

- Within each level, the **level mean places a constraint on the scores**
- One the level mean and all but one of the scores are known, the **final score is determined**
- Thus, for each level:

$$\text{degrees of freedom} = \text{number of scores} - 1$$

- The **total within-group degrees of freedom** is obtained by adding the degrees of freedom across all levels:

$$(N_{A_1} - 1) + (N_{A_2} - 1) + (N_{A_3} - 1)$$

Degrees Of Freedom

- Then for our worked example, which has 5 scores in each level:

$$\begin{aligned} &= (5 - 1) + (5 - 1) + (5 - 1) \\ &= 4 + 4 + 4 \end{aligned}$$

total within-group degrees of freedom = 12

Total Within-Group Variance

total within-group variance = $\frac{48}{\text{total within-group degrees of freedom}}$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Within-Group Variance

total within-group variance = $\frac{48}{12}$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Total Within-Group Variance

total within-group variance = $\frac{48}{12} = 4$

			
A_1 scores	A_2 scores	A_3 scores	
5	8	10	
7	8	11	
8	10	12	
9	11	13	
11	13	14	
$\bar{A}_1 = 8$	$\bar{A}_2 = 10$	$\bar{A}_3 = 12$	$\bar{Y} = 10$

Producing The F Statistic

The F Ratio

$$F = \frac{\text{between-group variance}}{\text{within-group variance}}$$

$$F = \frac{20}{4}$$

$$F = 5$$

Critical Values: F Table

Critical values of F ($\alpha = .05$)										
Denominator degrees of freedom (df_2)	Numerator degrees of freedom (df_1)									
	1	2	3	4	5	6	7	8	9	10
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.72	2.67
14	4.60	3.74	3.34	3.11	2.96	2.85	2.77	2.71	2.65	2.60
15	4.54	3.68	3.29	3.06	2.90	2.80	2.71	2.65	2.59	2.54
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.60	2.54	2.49
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.34

ANOVA Table

Source of variance	Sum of squares	Degrees of freedom	Variance (Mean square)	<i>F</i>
Between groups	40	2	20	5.00
Within groups	48	12	4	
Total	88	14		

Next Up, Assumptions Of ANOVA And Follow Up Procedures